

llmXive Automated Review of Perception or Prejudice: Can MLLMs Go Beyond First Impressions of Personality?

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The review focuses on the accuracy of factual claims and the support provided by citations and data within the manuscript.

1. Overstatement of “Un-grounded” Claims (Sec 1, Sec 5.1) The central claim of the paper is the “Prejudice Gap,” defined as “51% of correct ratings are not grounded in retrieved cues” (Abstract, Sec 5.1). This claim is derived from the Prejudice Rate (PR) metric, defined as $PR = \Pr[r_3 = 0 \mid r_1 = 1]$, where r_3 is the binary outcome of Task 3 (Grounding MCQs). The authors equate “failing the MCQ ($r_3 = 0$)” with “not being grounded.” This is a logical leap. A model might possess valid internal grounding (e.g., it “sees” the cue) but fail the specific MCQ due to distractor design, ambiguity in the question, or the specific phrasing of the options. By stating “not grounded” as a definitive fact based solely on MCQ failure, the paper conflates “failure to answer the probe correctly” with “absence of grounding capability.” The claim should be softened to “51% of correct ratings fail to retrieve the specific cue in the MCQ probe” to accurately reflect the evidence.

2. Inaccurate Summary Statistic (Sec 5.1) In Section 5.1, the text states: “Open-source Top-3 mean PR $\approx 47.0\%$.” Looking at Table 1 (Leaderboard), the Top-3 Open-source models listed are:

- Qwen3.5-397B: PR = 41.5%
- Qwen3-VL-235B: PR = 47.0%
- LLaVA-NeXT-7B: PR = 82.3% (Wait, the table shows LLaVA-NeXT-7B at the bottom with 82.3%. The “Top-3” usually implies the best performers. If we take the best 3 open models by HR or T3, we need to check the list. The table lists Qwen3.5, Qwen3-VL, and then skips to LLaVA-NeXT-7B at the bottom. The “Top-3” open models by HR are likely Qwen3.5 (15.9), Qwen3-VL (12.8), and perhaps another not explicitly highlighted as “Top-3” in the snippet but implied.

However, if we look at the PR values for the open models listed: 41.5, 47.0, 56.4 (Qwen3-VL-235B is 47.0, but the next one in the list might be different). Actually, looking at the provided text snippet for Table 1:

- Qwen3.5-397B: PR 41.5
- Qwen3-VL-235B: PR 47.0
- LLaVA-NeXT-7B: PR 82.3

If the “Top-3” refers to the best HR open models, the PRs might be different. But if the text claims the mean is 47.0, and the values are 41.5, 47.0, and 56.4 (assuming a third model with

56.4), the mean is $(41.5 + 47.0 + 56.4)/3 = 48.3$. If the third model is LLaVA (82.3), the mean is much higher. The value 47.0 appears to be the PR of the *second* best model, not the mean of the top three. This is a factual error in the summary statistic.

3. Regulatory Citation Context (Sec 1) The introduction cites the EU AI Act~\cite{council2024regulation} to support the claim that it “mandat[es] explainable evidence trails for high-risk systems.” While the Act does require transparency and human oversight, the specific requirement for “evidence trails” in the form of frame-accurate bounding boxes or specific cue retrieval for personality assessment is an interpretation by the authors, not a direct mandate of the regulation. The regulation focuses on risk management and transparency, not the specific technical implementation of “grounding” via MCQs. The phrasing suggests a direct regulatory requirement that may be overstated. It would be more accurate to say the Act “motivates the need for explainable evidence” rather than “mandating” this specific form of evidence.

4. Citation of “Funder’s Realistic Accuracy Model” (Sec 1, App A) The paper cites Funder (1995)~\cite{funder1995accuracy} to support the distinction between “perception” and “prejudice” and the use of “micro-cues.” This is a standard and appropriate citation in personality psychology. The claim that existing benchmarks fail to distinguish these is supported by the literature review. No factual error here, but the link between the psychological theory and the specific MCQ design (T3) is an operationalization choice, not a direct derivation from the citation.

Conclusion The paper makes a strong central claim about the “Prejudice Gap” that relies on equating MCQ failure with a lack of grounding. This is a methodological assumption that is stated as a fact. Additionally, there is a minor arithmetic error in the summary of the Open-source Top-3 PR. The regulatory claim is slightly overstated. These issues require minor revisions to the text to ensure the claims accurately reflect the evidence and calculations.

Action items.

- **[science]** The claim ‘51% of correct ratings are not grounded’ equates MCQ failure with lack of grounding. Failing a specific probe does not prove absence of internal grounding; the text should clarify this is ‘failure to retrieve the specific cue in the MCQ’.
- **[writing]** The EU AI Act citation implies a mandate for ‘evidence trails’ in the form of frame-accurate bounding boxes. The Act requires transparency, not this specific technical implementation. Rephrase to ‘motivates the need for explainable evidence’ rather than ‘mandating’.
- **[writing]** The text states the Open-source Top-3 mean PR is ~47.0%. Based on Table 1 values (41.5, 47.0, 56.4), the mean is 48.3%. The value 47.0 appears to be a single model’s score, not the mean. Correct the statistic or the model selection.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 is a clear and well-structured overview diagram that effectively visualizes the MM-OCEAN pipeline and benchmark tasks. The layout logically flows from input to processing to the three specific tasks, and the bottom ‘Demo’ section provides concrete examples that align perfectly with the caption’s description of the benchmark’s capabilities.

Figure 2

The figure provides a clear visual overview of the benchmark hierarchy and annotation density, but the ‘Per-video density’ table contains a metric (‘Trait analyses’) that conflicts with the task counts shown in the sunburst chart, and some outer ring labels are illegible due to tight spacing.

Figure 3

Figure 3 is a clear and well-structured diagram that effectively visualizes the five-stage annotation pipeline described in the caption. All stages, agents, and specific checks (C1-C4) are clearly labeled and logically connected, with no missing legends or illegible text.

Figure 4

The figure is a radar chart comparing model performance across categories, but it critically lacks a caption to define its content. Additionally, the title uses terminology (‘Closed’) that differs from the legend (‘Proprietary’), and the color-coding of axis labels is not explained.

Figure 5

The figure screenshot depicts a multiple-choice question interface rather than the annotation tool described in the caption, and it lacks the claimed bounding box overlays on the video frame.

Figure 6

The heatmap effectively visualizes the consistency of AI judges, but the caption uses ‘\$\$’ while the figure uses ‘Δ’ for the drop metric, creating a notation mismatch. Additionally, the colorbar label ‘Judge score (1-10)’ is potentially misleading regarding the scale of the drop values.

Figure 7

The figure effectively visualizes the performance gap across traits with clear data labels, but the redundant main title and the need to mentally invert accuracy to difficulty slightly hinder immediate communication.

Figure 8

The figure effectively visualizes the scaling trends but suffers from a critical design flaw where T2 values are multiplied by 10 to fit the same axis as T1/T3 percentages, making direct visual comparison of ‘performance’ misleading without careful inspection of the legend. The axis labels also slightly deviate from the caption’s terminology.

Figure 9

The bar chart effectively visualizes the performance gap between model subsets, but the scaling of the T2 metric to fit the ‘Score / Accuracy’ axis is not explicitly defined, and the color coding for delta values lacks a legend.

Figure 10

Figure 10 effectively visualizes the trade-off between accuracy and prejudice across models, but suffers from a confusing y-axis label, a typo in the x-axis label, and a lack of explanation for the ‘Trustworthy zone’ thresholds in the caption.

Figure 11

Figure 11 is a clear and well-structured heatmap that effectively visualizes per-model failure modes across four categories. The axes, units, and color scales are legible, and the caption accurately describes the content and sorting method.

Figure 12

The figure effectively visualizes rank reordering but contains a significant logical inconsistency where models with large rank changes are colored gray (stable) despite the caption defining stability as a change of less than 5 ranks.

Action items.

- **[science]** Figure 2(b): The ‘Per-video density’ table lists ‘Trait analyses / video’ as 5, but the sunburst chart (a) shows ‘Task 1 Personality Rating’ covers 1,104 videos. The table metric ‘Trait analyses’ is undefined and does not match the task count in the chart, creating a discrepancy in the reported data density.
- **[writing]** Figure 2(a): The outer ring labels for the ‘Visual Grounding’ and ‘Reasoning’ categories are rotated and tightly packed, making text like ‘TSInt 849q’ and ‘Spat 1007q’ difficult to read without zooming.
- **[fatal]** Figure 4: The figure has no caption (labeled ‘(no caption)’), making it impossible to verify what the radar chart represents or define the data series without guessing.
- **[science]** Figure 4: The chart title mentions ‘Top-3 Closed’ but the legend only defines ‘Top-3 Proprietary’ and ‘Top-3 Open-source’; the mapping between ‘Closed’ and ‘Proprietary’ is assumed but not explicitly stated in the figure or caption.

- **[writing]** Figure 4: The axis labels (e.g., ‘Personality Attribution’, ‘Counterfactual’) are colored to match the data series, but the legend does not explain this color-coding scheme, creating potential confusion about whether the colors represent categories or data series.
- **[science]** Figure 5: The caption claims the tool displays a ‘bbox overlay on the current frame,’ but the video player shows a person in a car with no visible bounding boxes or geometry overlays.
- **[science]** Figure 5: The caption describes a ‘Three-pane layout’ with specific annotation controls, but the screenshot shows a ‘MCQ Questions’ tab with multiple-choice options, which contradicts the described ‘atomic-cue list’ and ‘edit controls’ for bounding boxes.
- **[writing]** Figure 6: The title and caption use the symbol ‘\$\$’ to denote the drop metric, but the figure’s x-axis labels and colorbar use the symbol ‘ Δ ’ (Delta). This inconsistency between the rendered text and the caption description is confusing.
- **[writing]** Figure 6: The colorbar label ‘Judge score (1-10)’ is ambiguous; it is unclear if this scale applies to the raw T2 scores in the first two columns or the drop values in the third column, as the drop values (e.g., 1.22) fall outside the typical 1-10 range implied by the label.
- **[writing]** Figure 7: The main title ‘Neuroticism is the universally hardest Big Five trait’ is redundant with the caption and should be removed to reduce clutter.
- **[science]** Figure 7: The y-axis label ‘Mean T1 accuracy (%)’ is technically correct but the caption frames the figure as ‘difficulty’; adding a secondary label or note clarifying that lower accuracy equals higher difficulty would improve immediate interpretability.
- **[science]** Figure 8: The y-axis is labeled ‘Score / Accuracy’ but plots T2-Avg4 values scaled by 10 (e.g., 5.96 becomes ~60) alongside T1/T3 percentages (e.g., 51.3). This mixing of unscaled percentages and scaled raw scores on a single axis without explicit unit differentiation for each series is misleading and obscures the true magnitude of the T2 metric.
- **[writing]** Figure 8: The x-axis labels use inconsistent formatting for the parameter bands; the first group is labeled ‘8B’ while the caption describes it as ‘8B’, and the third group is ‘100B+’ while the caption uses ‘100B+’. While minor, the visual label ‘8B’ contradicts the caption’s ‘8B’ description.
- **[science]** Figure 9: The x-axis label ‘T2-Avg4 (1-10)’ indicates a 1-10 scale, but the y-axis is labeled ‘Score / Accuracy’ with a 0-70 range. This implies the T2 scores were scaled by 10 to match the axis, but this transformation is not explicitly stated in the caption or axis labels, making the direct visual comparison of bar heights misleading.
- **[writing]** Figure 9: The delta values (Δ) are color-coded (black vs. red) to indicate magnitude, but there is no legend or text in the caption explaining the threshold for this color change.
- **[science]** Figure 10: The ‘Trustworthy zone’ is defined in the plot as T1 $\geq$ 50% and PR $\geq$ 30%, but the caption ‘Right Rating, Wrong Cues’ and the title ‘Only 5 of 27 MLLMs reach the Trustworthy zone’ do not explain the specific criteria for ‘Trustworthy’ or the significance of the 50%/30% thresholds, leaving the reader to guess the benchmark’s success criteria.

- **[writing]** Figure 10: The y-axis label ‘Prejudice Rate, PR (%) ↓ better correct ratings without retrieved cues’ is grammatically confusing and cluttered; it should be split or rephrased to clearly indicate that lower PR is better and define what PR measures.
- **[writing]** Figure 10: The x-axis label ‘Task 1 accuracy (I%) → better’ contains a likely typo (‘I%’ instead of ‘%’), which reduces professionalism and clarity.
- **[science]** Figure 12: The legend defines ‘Drops 5’ and ‘Climbs 5’, but the plot contains many gray lines connecting points with rank differences clearly greater than 5 (e.g., Gemini 2.5 Flash moves from ~25 to ~4, a change of 21 ranks). This contradicts the caption’s definition of ‘Stable ($|\Delta| < 5$)’.
- **[writing]** Figure 12: The y-axis label ‘Rank (1 = best)’ is present, but the axis ticks are inverted (0 at top, 25 at bottom) without explicit indication, which can be confusing for readers expecting standard Cartesian coordinates.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on custom LaTeX macros and unexplained acronyms that hinder accessibility for non-specialist readers. In the Introduction, the term `\gapinv` is used to denote the “Prejudice Gap,” but the macro itself is opaque; the text should explicitly state “Prejudice Gap (PRG)” or similar on first mention rather than relying on a command. Similarly, in Section 4.2, the metric “RGM” (Rating-Grounding Misalignment) is introduced via Equation 4 without a textual definition of the acronym, forcing the reader to reverse-engineer the meaning from the formula or later context.

In the appendices, “TVD” is used in Table 3 without definition, and “TSJnt” appears in Table 5 as a category label for “Temporal-Spatial Joint,” which is unintuitive. The abbreviation “pp” for percentage points is used frequently in the results (e.g., Section 5.1) without clarification. While these terms are standard in specific sub-fields, the paper’s broad scope on MLLM safety and personality requires that every acronym be defined at first use and that custom commands be minimized in favor of plain English to ensure the “Prejudice Gap” concept is accessible to a wider audience.

Action items.

- **[writing]** The manuscript relies heavily on custom LaTeX macros and unexplained acronyms that hinder accessibility for non-specialist readers. In the Introduction, the term `\gapinv` is used to denote the “Prejudice Gap,” but the macro itself is opaque; the text should explicitly state “Prejudice Gap (PRG)” or similar on first mention rather than relying on a command. Similarly, in Section 4.2, the metric “RGM” (Rating-Grounding Misalignment) is introduced via Equation 4 without a textual definition of the

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a novel benchmark (MM-OCEAN) and a compelling high-level narrative distinguishing “perception” from “prejudice.” However, several core logical links between the proposed metrics and the final conclusions are tenuous or confounded.

First, the central claim of a “Prejudice Gap” (51% of correct ratings lack evidence) rests on the definition of the Prejudice Rate (PR). The paper defines PR as the probability that a model fails the grounding MCQ (T3) given it passed the rating task (T1). The logical leap here is equating “failing the MCQ” with “lacking evidence.” A model might correctly identify a cue in its open-ended reasoning (T2) but fail the specific MCQ due to distractor design, temporal granularity mismatches, or the specific phrasing of the question. Without a cross-validation showing that T2 reasoning *also* fails when T3 fails, the conclusion that the model is “prejudiced” (relying on pattern matching rather than cues) is not fully supported by the T3 metric alone.

Second, the significant performance gap between closed and open-source models on T3 (-26.6%) is attributed to “cue retrieval” capabilities. However, the paper acknowledges in Appendix A.9 that T3 accuracy is strongly negatively correlated with positional bias ($r \approx -0.68$). If open-source models generally exhibit higher positional bias (a known issue in many open weights), the observed gap may be an artifact of this bias rather than a fundamental difference in visual grounding or cue retrieval. The causal claim that open models are worse at “retrieving cues” is not robustly supported without controlling for this confounding variable.

Third, the analysis in Appendix A.8 attempts to isolate the effect of “reasoning capability” by comparing a “reasoning-capable” subset against a “non-reasoning” subset. The authors explicitly state this comparison is “Confounded by size/family/generation.” Logically, if the groups differ in size and architecture, one cannot validly conclude that the observed performance difference (+18.3pp on T3) is caused by “reasoning capability.” The conclusion drawn from this observational data is unsupported.

Finally, the definition of “Holistic-grounding Rate” (HR) requires success in T1, T2, and T3. The paper claims HR discriminates models better than single tasks. While the variance (CV) is high, the logical consistency of HR as a metric depends on the independence of T1, T2, and T3. If T2 (AI-as-Judge) is heavily influenced by T1 correctness (as noted in the “confidently-wrong” check), the metric may be circular. The paper notes the judge penalizes wrong T1, but the extent to which T2 scores are *driven* by T1 rather than independent reasoning needs to be clearer to validate HR as a distinct measure of “holistic” grounding.

These issues do not invalidate the benchmark’s utility but require the authors to temper their causal claims regarding “prejudice” and “reasoning capability” or provide additional ablation studies to rule out the identified confounders.

Action items.

- **[writing]** The paper presents a novel benchmark (MM-OCEAN) and a compelling high-level narrative distinguishing “perception” from “prejudice.” However, several core logical links between the proposed metrics and the final conclusions are tenuous or confounded. First, the central claim of a “Prejudice Gap” (51% of correct ratings lack evidence) rests on the definition of the Prejudice Rate (PR). The paper defines PR as the probability that a model fails the grounding MCQ (T3) given it passed the rating task

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes strong claims regarding the “Prejudice Gap” and MLLM failures in grounding personality ratings. While the benchmark is rigorous, the interpretation occasionally extrapolates beyond the data’s specific constraints.

First, the central statistic that “51% of correct ratings are not grounded” (Abstract, Section 5.1) is derived from the Prejudice Rate (PR) metric, which depends on specific thresholds ($\theta_1 = 0.5, \theta_2 = 0.7, \theta_3 = 0.5$). Although Appendix F shows model *rankings* are stable across threshold variations, the paper does not quantify how the absolute “51%” figure shifts if these cutoffs change. Presenting this as a fixed, pervasive statistic without a sensitivity range for the metric itself is an over-claim. The authors should clarify that this figure is contingent on the specific threshold choices.

Second, the claim that proprietary and open models differ “significantly” in cue retrieval with a “-26.6% gap” (Section 5.1, Conclusion) is based on the mean performance of the Top-3 models in each category. The paper does not report a statistical significance test (e.g., a t-test or confidence intervals) for the difference between these two small groups ($n = 3$). Given the variance in model performance and the small sample size, asserting a “significant” difference is statistically premature. The data supports a *large observed gap*, but the statistical significance of this gap between the two populations is not established.

Finally, the conclusion characterizes the Prejudice Gap as “pervasive” (Abstract, Conclusion). While the data supports this within the specific context of the MM-OCEAN benchmark (derived from ChaLearn First Impressions V2), the dataset is limited to short, single-speaker, English videos with a known Western cultural bias (acknowledged in Appendix Ethics). Generalizing this failure mode as a universal trait of MLLMs in “human-centric roles” (Introduction) overreaches the scope of the data. The paper should temper its conclusion to reflect that this gap is pervasive *within the specific constraints of short-form, Western-centric video clips* rather than as a general law of MLLM behavior.

Action items.

- **[writing]** The ‘51% ungrounded’ claim (Abstract) relies on fixed thresholds. Provide a sensitivity range for this metric or clarify it is threshold-dependent to avoid over-claiming a

fixed statistic.

- **[science]** The ‘significant’ -26.6% gap between Top-3 closed/open models (Sec 5.1) lacks a statistical significance test (e.g., t-test) for n=3 groups. Replace ‘significant’ with ‘large observed’ or add statistical validation.
- **[writing]** The ‘pervasive’ Prejudice Gap claim (Conclusion) overgeneralizes from Western-centric, short-video data. Qualify the scope to the dataset’s specific constraints in the main conclusion.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript addresses a critical safety concern: the distinction between genuine perception and algorithmic prejudice in personality assessment. The “Prejudice Gap” metric is a valuable contribution to safety evaluation. However, several ethical and safety gaps require attention before publication.

First, the **Ethics and Responsible Use** section (Appendix E) is currently too brief regarding the specific risks of the new annotations. While the authors acknowledge the Western bias inherited from ChaLearn First Impressions V2, they do not detail how the 24 human annotators or the “Psychologist” stage mitigated cultural bias when generating the 5,320 new MCQs and trait analyses. Without a statement on the demographic diversity of the annotators or a cross-cultural validation step, the benchmark risks encoding and amplifying specific cultural stereotypes under the guise of “grounded” reasoning.

Second, the **dual-use risk** is significant. The paper explicitly mentions “interview screening” as a deployment scenario in the Introduction. Although the authors state the benchmark is a “diagnostic research tool,” the detailed “Prejudice Gap” and “Holistic-grounding Rate” metrics could be misinterpreted by industry actors as a validation of automated personality screening. The Ethics section must include a stronger, explicit prohibition against using MM-OCEAN for real-world hiring, lending, or surveillance decisions without rigorous, independent fairness audits and human-in-the-loop oversight.

Third, **data privacy and consent** require clarification. The dataset consists of 1,104 video clips of individuals. While the source is public, the new annotations (atomic behavioral cues, bounding boxes, and specific trait scores) create a much more granular profile of the subjects than the original dataset. The authors must clarify the consent status of the original subjects regarding this specific secondary use for personality profiling and ensure the CC-BY-NC license explicitly covers the derived personality labels to prevent misuse.

Finally, the **AI-as-Judge** protocol (Appendix) uses GPT-4o-mini to score reasoning. While cross-judge robustness is tested, the potential for the judge model to inherit its own biases

regarding “appropriate” personality expressions (e.g., penalizing non-Western communication styles) is not discussed. A brief analysis of the judge’s own cultural bias would strengthen the safety claims.

Action items.

- **[writing]** The manuscript addresses a critical safety concern: the distinction between genuine perception and algorithmic prejudice in personality assessment. The “Prejudice Gap” metric is a valuable contribution to safety evaluation. However, several ethical and safety gaps require attention before publication. First, the Ethics and Responsible Use section (Appendix E) is currently too brief regarding the specific risks of the new annotations. While the authors acknowledge the Western bias inherited from

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a substantial dataset (MM-OCEAN) and a novel evaluation framework distinguishing between correct personality ratings and grounded reasoning. The sample size (1,104 videos, 5,320 MCQs) is robust for benchmarking, and the multi-stage human annotation pipeline (24 annotators, 77% agreement) provides a solid foundation for the ground truth. The distinction between “perception” and “prejudice” is operationally defined through the Prejudice Rate (PR) and Holistic-grounding Rate (HR), which are mathematically sound given the three-task setup.

However, the scientific evidence supporting the causal interpretation of the “Reasoning Capability” effect is weak. In Appendix E002 (Table:reasoning), the authors compare “Reasoning-capable” vs. “Non-reasoning” models, finding a +18.3pp gap in T3. The text explicitly admits this is “Confounded by size/family/generation.” Without a controlled experiment (e.g., ablating reasoning chains in a single model family) or a multivariate regression controlling for parameter count and training data, the claim that “reasoning capability” is the driver of the gap is speculative. The data equally supports the hypothesis that larger, more capable models simply have better visual grounding architectures, regardless of their “reasoning” label.

Additionally, the robustness of the central “Prejudice Gap” claim (51% of correct ratings are ungrounded) depends heavily on the validity of the T3 MCQs as a proxy for “evidence.” The paper notes that 153 questions (1.8%) were answered correctly by no model, and difficulty varies widely (30.7% to 64.8%). If the MCQs are poorly constructed or if the “correct” answer relies on subtle cues not captured by the distractors, the PR metric may overstate the “prejudice.” A sensitivity analysis showing how PR fluctuates when excluding the most difficult or ambiguous items is necessary to confirm the stability of the 51% figure.

Finally, while the cross-judge correlation is high ($\rho=0.94$), the GPT-4o-mini judge exhibits a consistent +1.0 point self-preference bias (Table:cross_judge). Since the HR metric requires

passing a threshold ($\theta_2=0.7$) on the T2 score, this bias could artificially inflate the HR for GPT-4o-mini family models. The authors should demonstrate that the ranking of models by HR remains stable if the T2 scores are adjusted to remove this systematic bias.

Action items.

- **[science]** The ‘Reasoning Capability’ analysis (Appendix E002, Tab:reasoning) is observational and explicitly confounded by model size and family. The claim that reasoning capabilities drive the HR/PR gap requires a controlled ablation (e.g., same base model with/without RL) or a regression analysis controlling for parameters to rule out size as the primary driver.
- **[science]** The ‘Prejudice Gap’ (PR) metric relies on the assumption that T3 MCQs perfectly capture the evidence used for T1 ratings. The paper notes 153 ‘human-only’ questions (1.8%) and varying difficulty. A sensitivity analysis is needed to show that the 51% PR statistic is robust to the inclusion/exclusion of these ambiguous or extremely difficult items, rather than being an artifact of the specific MCQ distractors.
- **[science]** The AI-as-Judge for Task 2 (T2) shows a self-preference bias of +1.0 point for GPT-4o-mini (Tab:cross_judge). While the authors argue this doesn’t distort ranking, the absolute score inflation could impact the calculation of the ‘Holistic-grounding Rate’ (HR) if the T2 threshold ($\theta_2=0.7$) is sensitive to this bias. Report HR sensitivity to a ± 1.0 score shift.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical analysis in the manuscript is generally sound in its design of metrics (PR, HR, RGM) but lacks necessary inferential statistics to support the strength of its central claims.

1. Lack of Uncertainty Quantification: The paper relies heavily on point estimates for aggregate metrics. For instance, Section 5.1 states the mean Prejudice Rate is 51.3% and the mean HR is 10.4%. Without confidence intervals (CIs) or standard errors, it is impossible to determine if the observed “Prejudice Gap” is statistically robust or if it could be an artifact of the specific 27 models sampled. Similarly, the “Closed vs. Open” gap (14.5% vs 47.0%) is presented as a definitive finding. A formal hypothesis test (e.g., independent samples t-test or Mann-Whitney U test) with a reported p-value is required to validate that this difference is not due to random variation.

2. Correlation Reporting: In Appendix A.10 (Positional Bias), the authors report a correlation of $r \approx -0.68$ between positional bias and T3 accuracy. The manuscript does not specify whether this is Pearson or Spearman correlation, nor does it provide a p-value. Given the non-linear nature of model performance and the potential for outliers, a non-parametric test

(Spearman) is likely more appropriate, but the significance level must be explicitly stated to support the claim that “every model with $\sigma > 10$ ranks in the bottom third.”

3. AI-as-Judge Robustness: The cross-judge robustness analysis in Appendix A.9 uses a subset of 200 videos to calculate Spearman rank correlations ($\rho = 0.94$ and 0.92). While the high correlation is promising, the sample size is relatively small for establishing generalizability. The authors should report the 95% confidence intervals for these correlation coefficients. Without CIs, the precision of the estimate is unclear, and the claim that the ranking is “not an artifact” is statistically weaker than it could be.

4. Multiple Comparisons: The paper evaluates 27 models across multiple tasks and metrics (T1, T2, T3, HR, PR, CR, IR, RGM). While the primary focus is on the “Prejudice Gap,” the extensive reporting of per-category accuracies (Appendix A.7) and per-trait difficulties (Appendix A.6) invites multiple comparison issues. The authors should acknowledge this and, if any specific pairwise comparisons between models or categories are highlighted as “significant,” they should ideally be corrected (e.g., Bonferroni or Benjamini-Hochberg) or at least discussed in the context of Type I error inflation.

Recommendation: The authors should re-run their analysis to include confidence intervals for all mean metrics and p-values for all comparative claims (Closed vs. Open, Correlation tests). This will significantly strengthen the statistical validity of the “Prejudice Gap” conclusion.

Action items.

- **[science]** Report confidence intervals or standard errors for all aggregate metrics (e.g., mean PR, HR, T3 accuracy) in Table 1 and Section 5.1. Point estimates alone do not convey the stability of the ‘Prejudice Gap’ claim across the 27 models.
- **[science]** Clarify the statistical test used to validate the ‘Closed vs. Open’ gap (Section 5.1). A simple difference of means (14.5% vs 47.0%) is insufficient; provide a p-value or effect size (e.g., Cohen’s d) from a t-test or non-parametric equivalent to support the claim of a ‘significant’ gap.
- **[science]** In Section 5.2 and Appendix A.10, the correlation between positional bias and T3 accuracy is reported as $r = -0.68$. Specify if this is Pearson or Spearman correlation and provide the associated p-value to confirm statistical significance.
- **[science]** The AI-as-Judge robustness analysis (Appendix A.9) reports Spearman correlations ($\rho = 0.94, 0.92$) but lacks confidence intervals for these correlation coefficients. Given the small sample size ($n=200$ for the subset), CIs are necessary to assess the precision of the rank stability claim.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript presents a compelling benchmark with a clear narrative arc, effectively distinguishing between “perception” and “prejudice” in MLLMs. The writing is generally concise and technical. However, several structural and formatting inconsistencies hinder readability and professional polish.

First, there is significant redundancy between the main text and the appendices. Tables defining the MCQ taxonomy (Table 2 in main, Table 2 in Appendix) and the Leaderboard (Table 3 in main, Table 1 in Appendix) are duplicated. While appendices often contain extended data, repeating full tables with identical content disrupts the reading flow. The authors should consolidate these, keeping the primary tables in the main text and moving only extended breakdowns or raw data to the appendix.

Second, the handling of figure references and definitions requires attention. The main text introduces Figures 3 and 4 (radar and archetypes), but the appendix also contains definitions for figures with similar themes or duplicate labels (e.g., `fig:judge_reliability` appears in both the main text and Appendix). This risks compilation errors or broken cross-references. A systematic check of all `\label` and `\ref` commands is necessary to ensure uniqueness and correct mapping.

Third, the “Code and Data Availability” section is underdeveloped. It currently states only “Available in project repository,” which is insufficient for a published preprint. Given the critical elements list includes specific URLs (e.g., <https://github.com/kkkcx/MM-OCEAN>), these should be explicitly cited in the main text to ensure immediate accessibility for readers.

Finally, minor grammatical polish is needed in the Appendix. For instance, the “Human Annotation Protocol” section uses fragmented sentences (“24 trained annotators performed Stage 1 verification”) that could be smoothed into full prose for better cohesion. Addressing these structural and stylistic issues will significantly enhance the manuscript’s clarity and readiness for publication.

Action items.

- **[writing]** The manuscript contains duplicate content across the main body and appendices. Specifically, Table 1 (Leaderboard) and Table 2 (MCQ Taxonomy) appear in both Section 4 and the Appendix with slight formatting variations. Consolidate these into the main text or the appendix to improve flow and reduce redundancy.
- **[writing]** In Section 5.2 (Diagnostics), the text references ‘Figure 3’ and ‘Figure 4’ for radar and archetype plots, but the LaTeX source defines them as Figure 3 and Figure 4 in the main body. However, the Appendix contains duplicate figure definitions (e.g., Fig: judge_reliability) that may cause compilation conflicts or confusion. Ensure unique label definitions and consistent cross-referencing throughout.
- **[writing]** The ‘Code and Data Availability’ section (end of main text) is extremely brief (‘Available in project repository’) and lacks the specific URLs provided in the critical elements list (e.g., GitHub/HuggingFace links). Expand this section to include direct links for reproducibility.